UNIVERSITY OF WATERLOO

GIS in Hydrology

Core Concepts, Spatial Operations & HRU Discretization

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Download the lecture slides and lab materials from LEARN or GitHub (scan the QR code below).

Outline

- **What Is GIS & Why GIS Matters in Hydrology**
- **Core GIS Concepts**
- **Key GIS Operations for Hydrology**
- **ArcGIS Pro Preview**
- **Lab Preview: HRU Discretization Workflow**

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What Is a Geographic Information System?

Geographic Information System (GIS): *from 1960s*

- A computer system to **capture, store, analyze, and visualize** data **tied to locations**.
- GIS is a toolbox for answering **location-based** questions with data.
- It is **NOT** just making maps!

Geographic (Where)	**Location**: coordinates, geometry (point, line, polygon) **Spatial Reference**: Coordinate Reference System (CRS), projection, units **Spatial relationship**: distance, adjacency, containment, networks
Information (What)	**Attributes**: tables, fields, IDs, metadata **Measurements**: precipitation, temperature, land cover, soil class, elevation **Derived information**: flow direction, flow accumulation, zonal statistics
System (Workflow)	**Tools and workflow**: collect → store → process → analyze → visualize → share **Data management**: geodatabase, file system, reproducibility, versioning **People**: assumptions, QAQC, decision-making

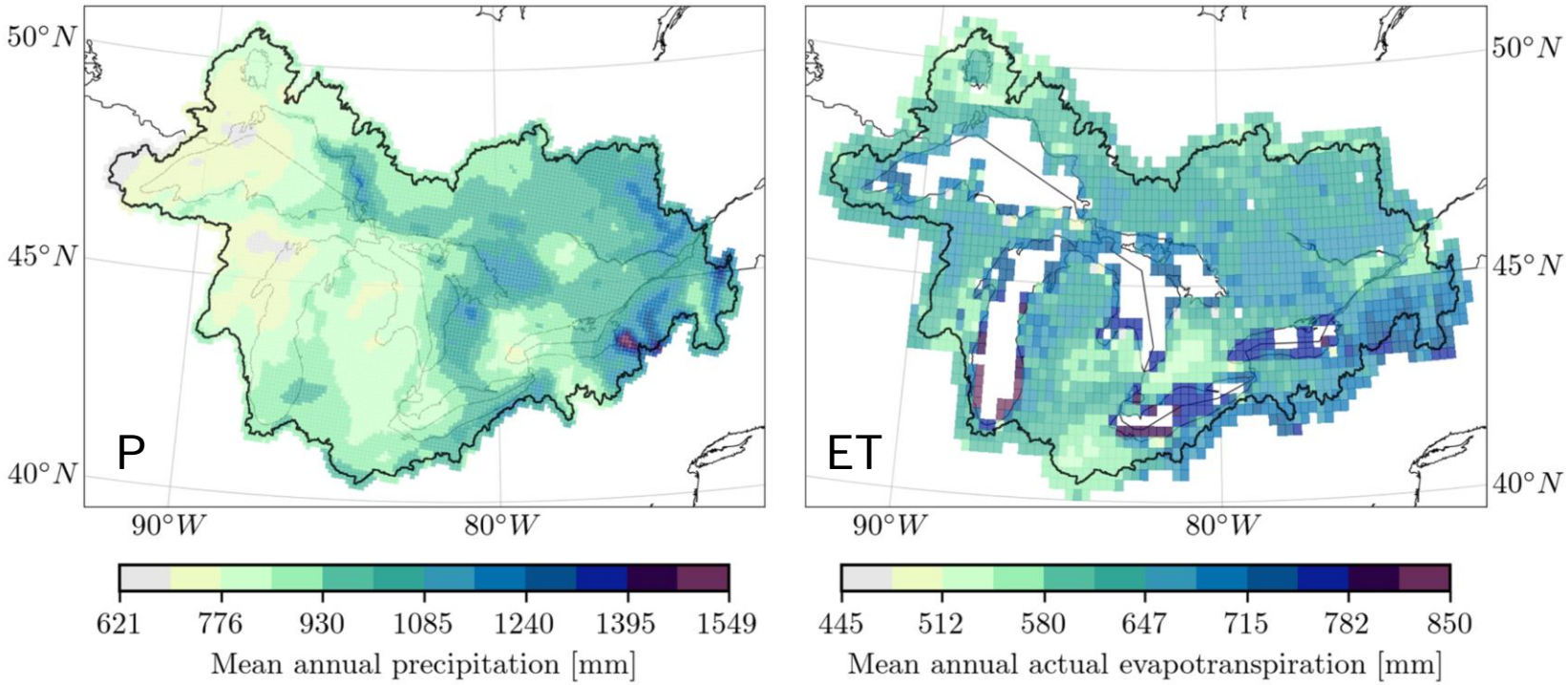
What Is a Geographic Information System?

GIS = linking **Where** (Geographic) + **What** (Information) through a reproducible **Workflow** (System)

Example GIS work:
To analyze spatial variability of mean annual P & ET in Great Lakes Region.

1) Process P & ET data everywhere
2) Link annual P & ET to locations
3) Visualize them in a map
4) Assess the spatial variability

The right-side figure does not show GIS workflows behind it.

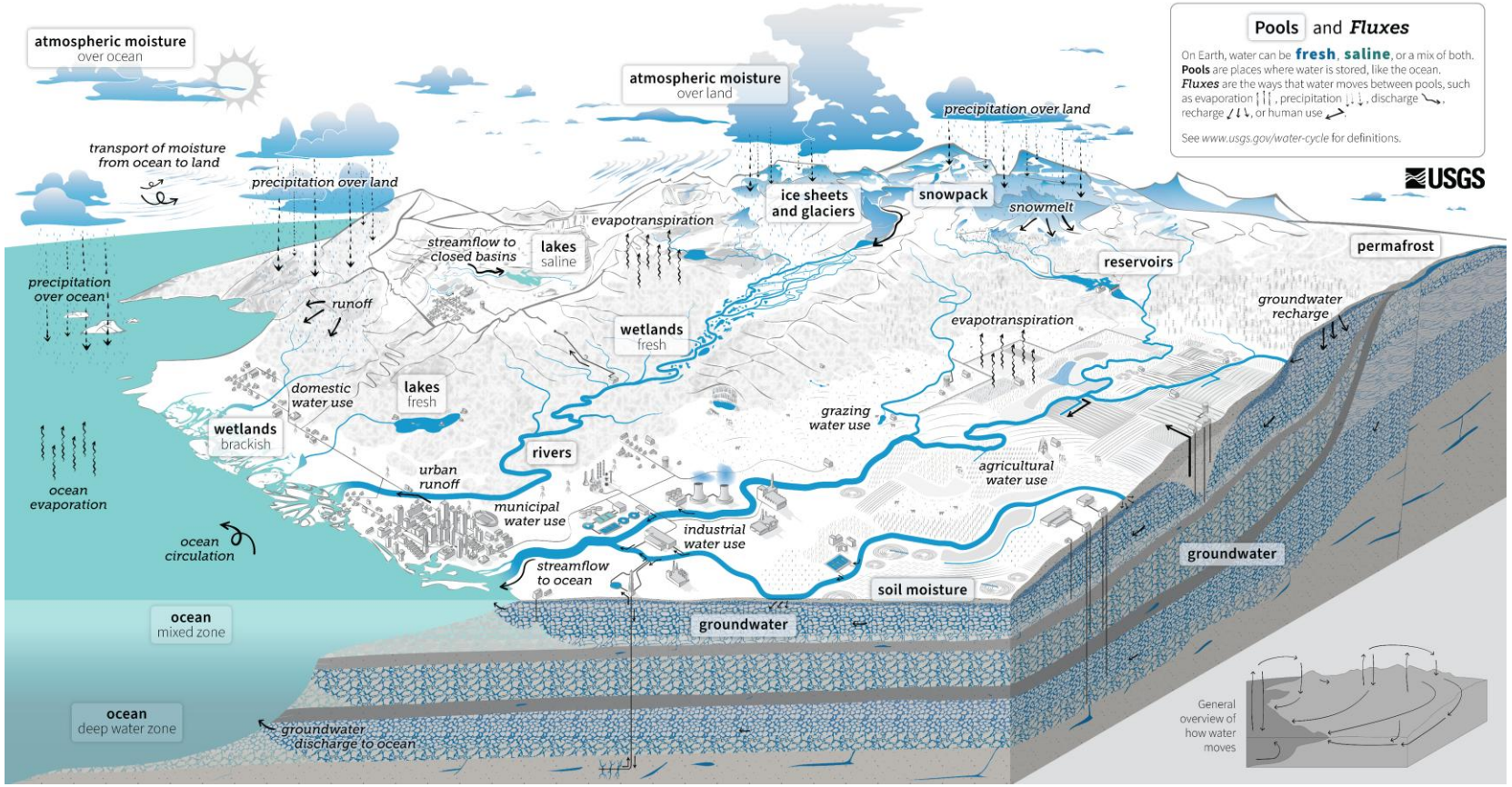


(Mai et al., HESS, 2022)

Mai, J., Shen, H., Tolson, B. A., Gaborit, É., Arsenault, R., Craig, J. R., et al. (2022). The Great Lakes Runoff Intercomparison Project Phase 4: the Great Lakes (GRIP-GL). Hydrology and Earth System Sciences, 26(13), 3537–3572. https://doi.org/10.5194/hess-26-3537-2022

Why GIS Matters in Hydrology

- Hydrologic processes are **inherently spatial** and vary across the landscape.
- Key spatial controls: **meteorology, topography, geology/soils, and drainage networks**.
- Hydrologic analysis integrates **multi-source datasets**: gridded climate, remote sensing, and **point- to basin-scale data** (e.g., gauges, lakes/reservoirs, land use/irrigation, and human influences such as agriculture, industry, and municipalities).



(USGS, The Water Cycle)

GIS helps integrate multi-source data, represent spatial heterogeneity, and enable semi-distributed and distributed hydrologic modeling.

Why GIS Matters in Hydrology

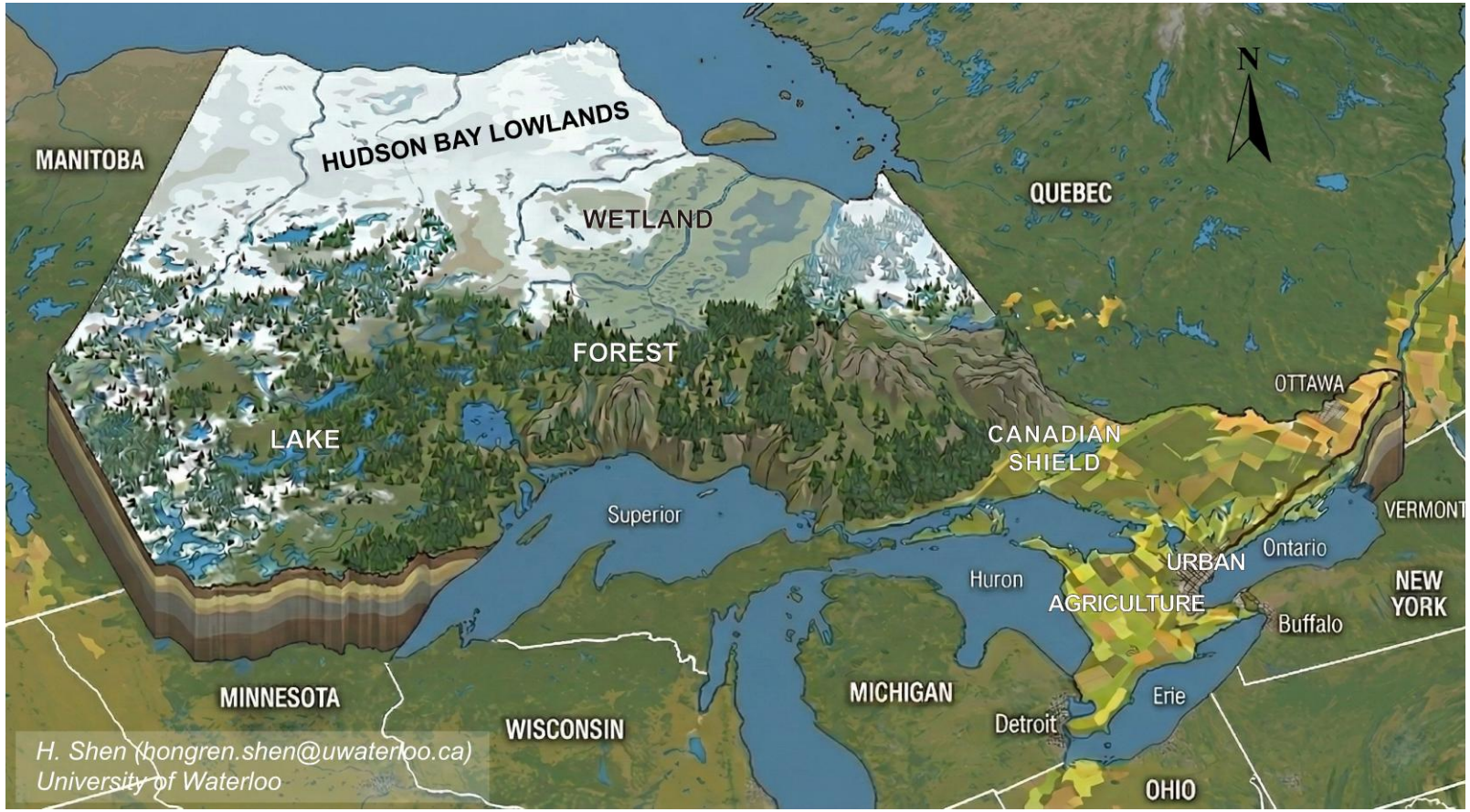
Lumped models vs. Distributed models

- Input met. data (forcing)
- Lake and river network
- Landscape representation
- Modeling unit (single vs. HRUs)
- Large basins vs. small basins

What is an HRU?

- Hydrologic Response Unit (HRU)
- Similar hydrologic behavior
- Land cover × Geology × Topography
- Represent heterogeneity in runoff generation and storage

Ontario Landscape Heterogeneity



Core GIS Concepts

(1) Vector vs. Raster

(2) Attribute tables + IDs (for vector data)

(3) Coordinate Reference System (CRS)

(4) Resolution, extent and scale

(5) Spatial relationships

(6) Overlay concept

Core GIS Concepts

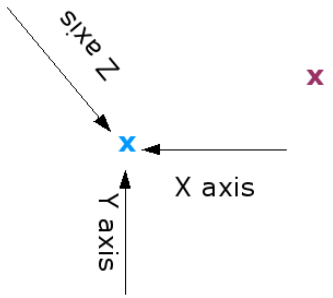
(1) Vector vs. Raster

Vector data: points, lines, polygons (discrete features)

- A vector feature has its **geometry** (shape) and **attributes**
- Examples: hydrometric gauges, rivers, subbasins/HRUs

Vector Point Feature

Point Geometry (indicates the x,y and z position of the feature)



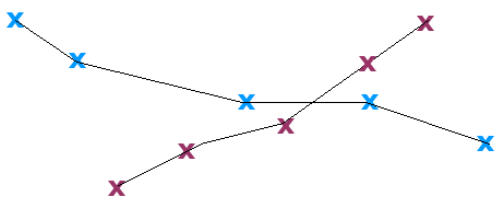
Point attributes (describe the feature)

Id, Name, Description

1, Tree, Outside our classroom
2, Light post, At the school entrance

Vector Polyline Feature

Polyline Geometry (a series of connected vertices that do not form an enclosed shape)



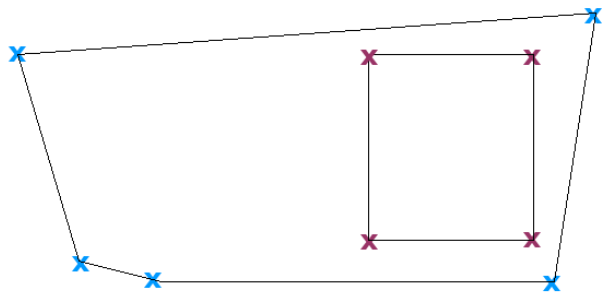
Polyline attributes (describe the feature)

Id, Name, Description

1, Footpath 1, From class to the playground
2, Footpath 2, From the school gate to the hall

Vector Polygon Feature

Polygon Geometry (a series of connected vertices that do form an enclosed shape)



Polygon attributes (describe the feature)

Id, Name, Description

1, School Boundary, Fenceline for the school
2, Sports Field, We play soccer here

(QGIS v3.40 Documentation)

Core GIS Concepts

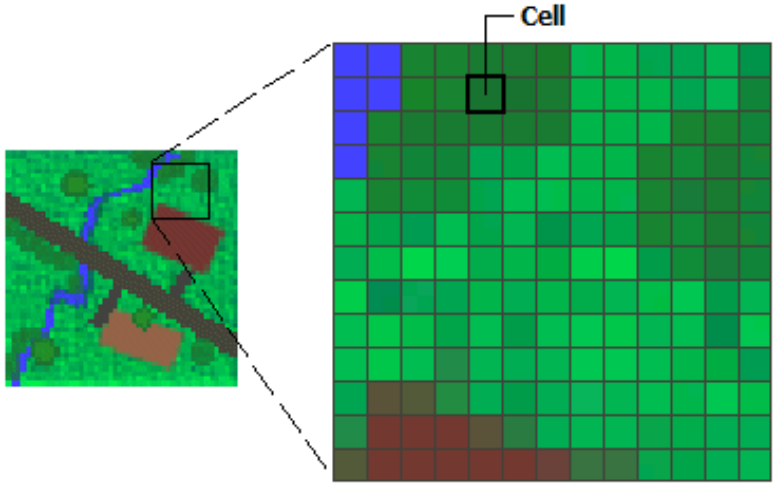
Vectors are not good at representing features that are not homogeneous (entirely the same) all over, e.g., meteorological variables

(1) Vector vs. Raster

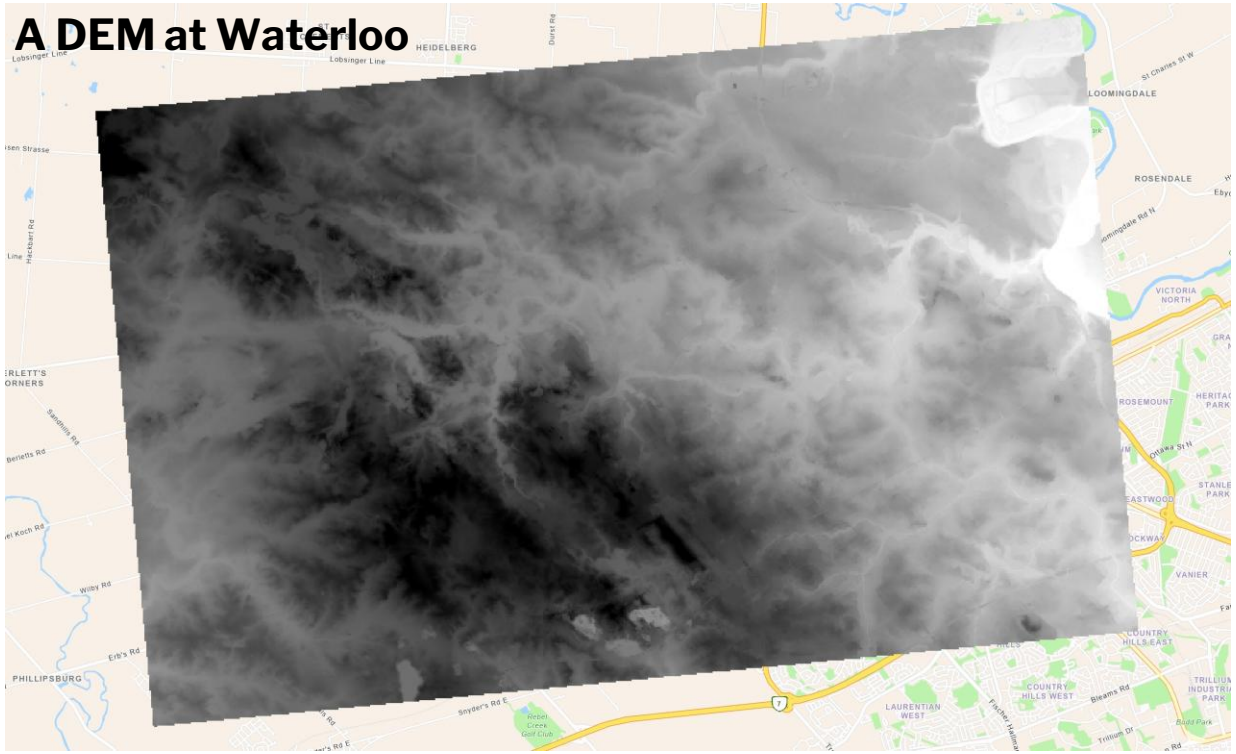
Raster: **grid-cell data** (continuous or categorical)

- Good for representing data changes continuously across surface.
- Examples: Digital Elevation Model (DEM), land cover, gridded climate forcing

Raster data defined by cells (pixels)



A DEM at Waterloo

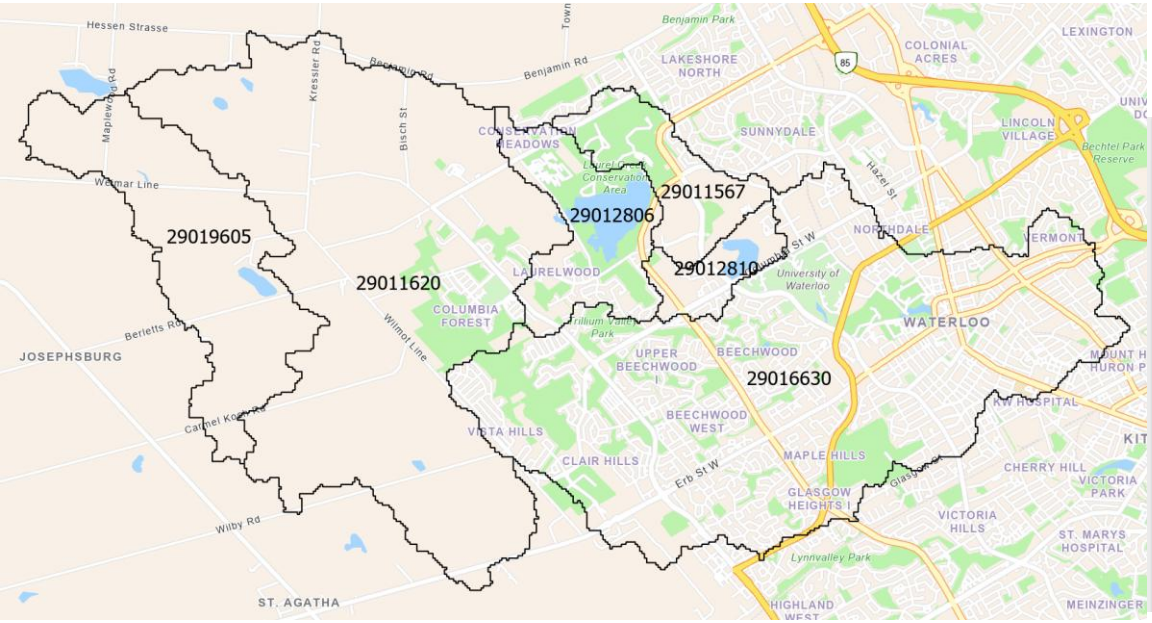


Core GIS Concepts

(2) Attribute tables + IDs (for vector data)

- Attributes for a vector feature are stored in a **table**. A table is like a spreadsheet.
- Each column in the table is called a **field**.
- Each row in the table is a **record**.

Example subbasin polygon feature and its attribute table (SubId displayed on map)



FID: Feature ID is an internal row index in GIS software.
Shape: Feature geometry type.

finalcat_info_v2-0-1

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete

	FID	Shape	SubId	DowSubId	RivSlope	RivLength	BasSlope	BasAspect	BasArea	BkfWidth	BkfDepth
1	0	Polygon	29011567	29012810	0.003382	1932.792206	2.068323	153.404199	2595600	7.092465	0.267573
2	1	Polygon	29011620	29012806	0.002689	5728.965363	3.262719	166.917507	20571300	6.165315	0.246001
3	2	Polygon	29012806	29011567	-1.2345	-1.2345	1.757881	160.431606	3523500	6.559344	0.255317
4	3	Polygon	29012810	29016630	-1.2345	-1.2345	1.747721	161.970201	1320300	6.946745	0.264261
5	4	Polygon	29016630	-1	0.003011	5442.274888	2.289449	162.46897	23256000	8.957982	0.307815
6	5	Polygon	29019605	29011620	-1.2345	-1.2345	3.490921	167.239563	6923700	3.37832	0.171469

Core GIS Concepts

(3) Coordinate Reference System (CRS)

ALWAYS CHECK THE CRS OF YOUR INPUT DATA!

Do geoprocessing in a **projected CRS (meters)**

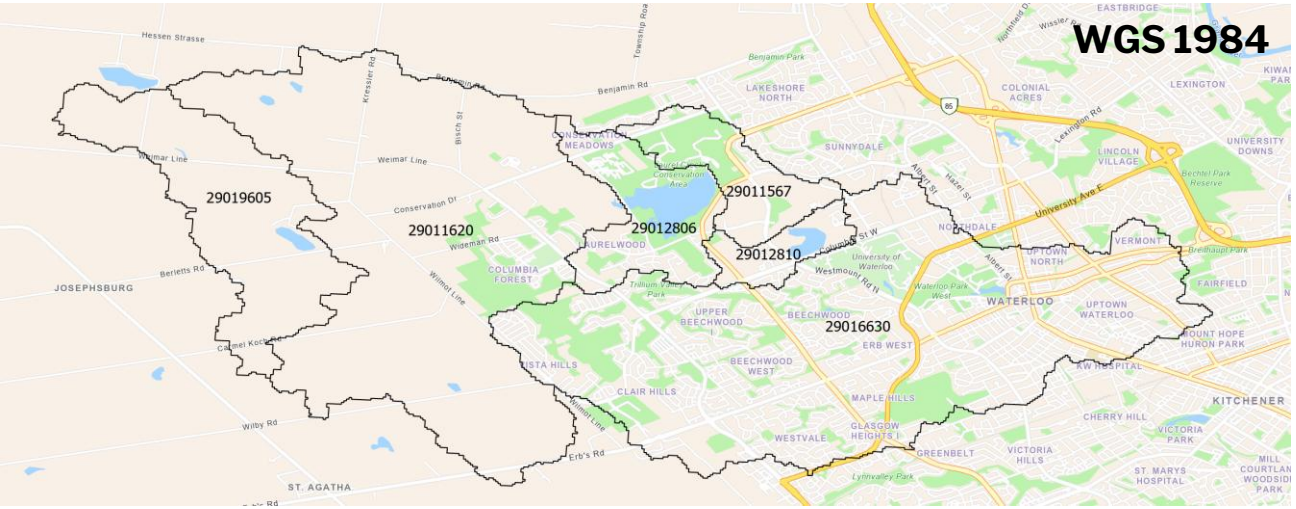
Keep a consistent CRS across layers

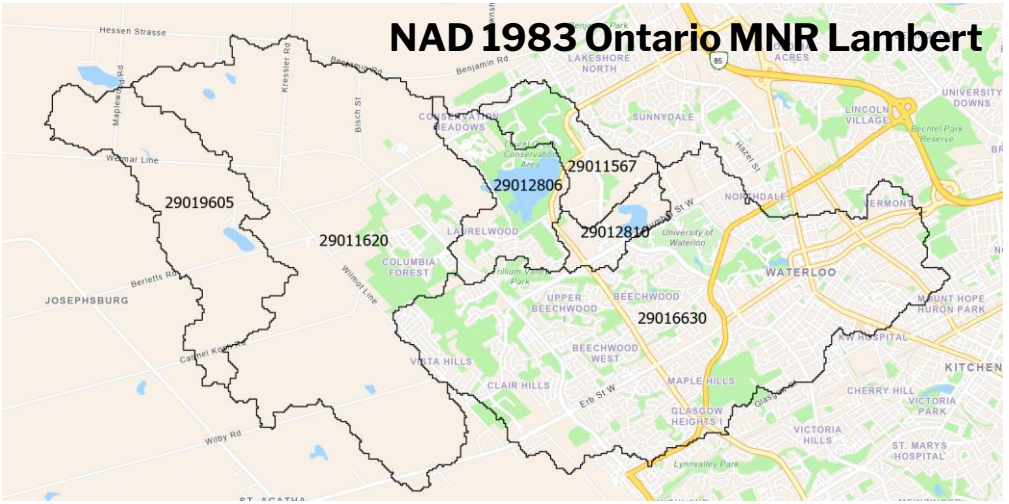
Geographic Coordinate Systems

- Degrees (lat/long) : angular units, not linear distance
- Example: WGS 1984, NAD 1983
- Good for: global maps, GPS coordinates, data sharing
- Not for calculation: area/length/slope/buffer in degrees will be wrong

Projected Coordinate Reference Systems

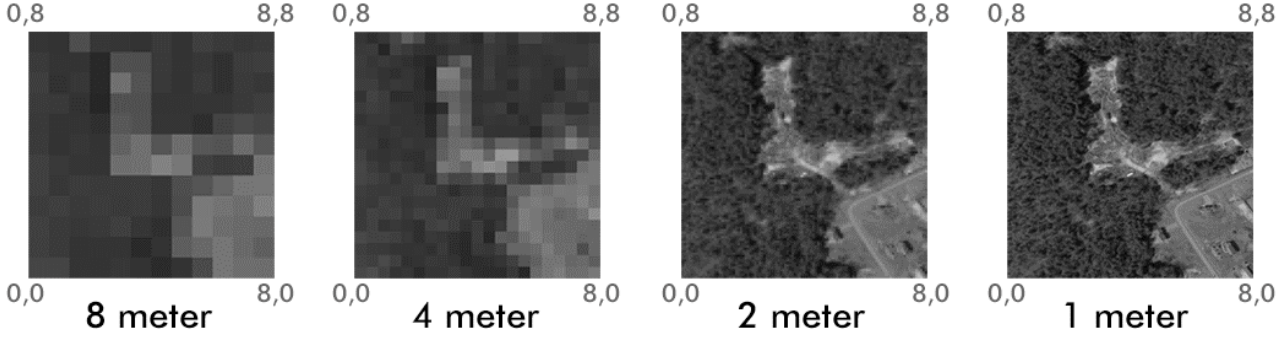
- Meters or feet: linear units on a plane
- Example: NAD 1983 Ontario MNR Lambert
- Use for: area, distance, buffering, river length, DEM-derived slope





Core GIS Concepts

(4) Resolution, extent and scale



Raster across same extent but different resolutions

Raster resolution

- Raster resolution depends on **cell size** (e.g., 30 m × 30 m, 1 km × 1km)
- Smaller cells (**higher resolution**) capture more details but increase noise, size and computation
- Larger cells (**lower resolution**) smooth variability but can miss small features

Extent

- Extent (spatial coverage) is critical when multiple layers are considered in one study
- Mismatched extent leads to **missing data or NoData** edges

Scale

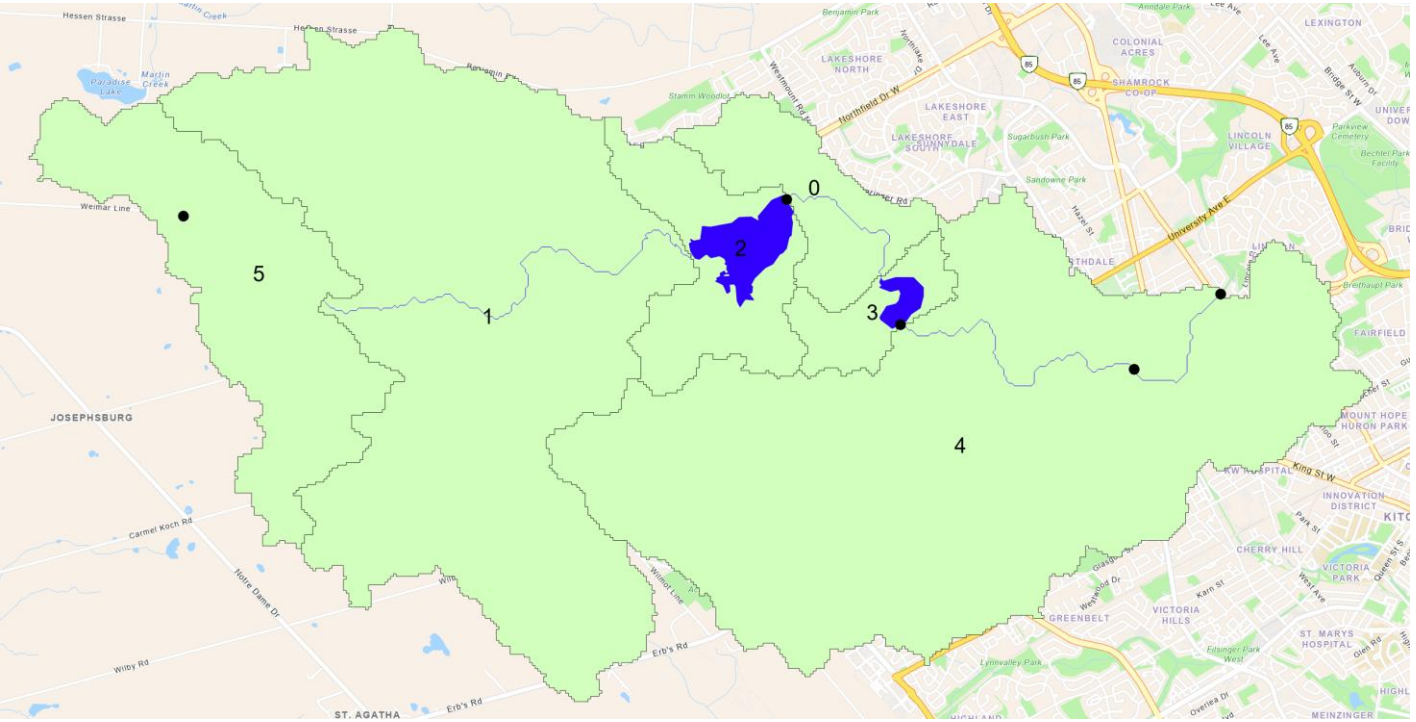
- Scale is a factor related to GIS operations.
- Is finer scale always better???

e.g., Suppose we have all 30 meters input data, does it make sense to create 5 meters HRUs?

Core GIS Concepts

(5) Spatial relationships

- Spatial relationships describe how geographic objects relate in space: within, contains, intersects, touches, nearest, upstream/downstream
- Basis of most GIS queries and joins



<u>Common relationships (in hydrology)</u>

- **Within / Contains**: Which subbasin contains a gauge? Which HRUs are within a subbasin?
- **Intersect / Overlap**: Which soil polygons intersect a subbasin? Where do rivers cross wetlands?
- **Touch / Adjacent**: Which HRUs share boundaries?
- **Nearest**: Nearest gauge to a dam point; nearest river reach to an outlet
- **Connectivity (network)**: Upstream vs downstream relationships in river networks

Core GIS Concepts

(6) Overlay concept

- Overlay combines two or more spatial layers to produce a new layer where features are split by their overlaps.
- Result: each new feature inherits attributes from all input layers → new analysis units.
- Typical example: HRUs are typically created by overlaying multiple key layers

e.g., **Subbasins × Land cover × Vegetation × Soils/Geology × Slope/Elevation zones**

Overlay turns "geospatial layers" into "units for modeling" with area fractions and parameters.

Vector overlay (polygons/lines/points)

- Tools: **Intersect, Union, Identity, Spatial Join**
- Output: new polygons (often many with **slivers**)

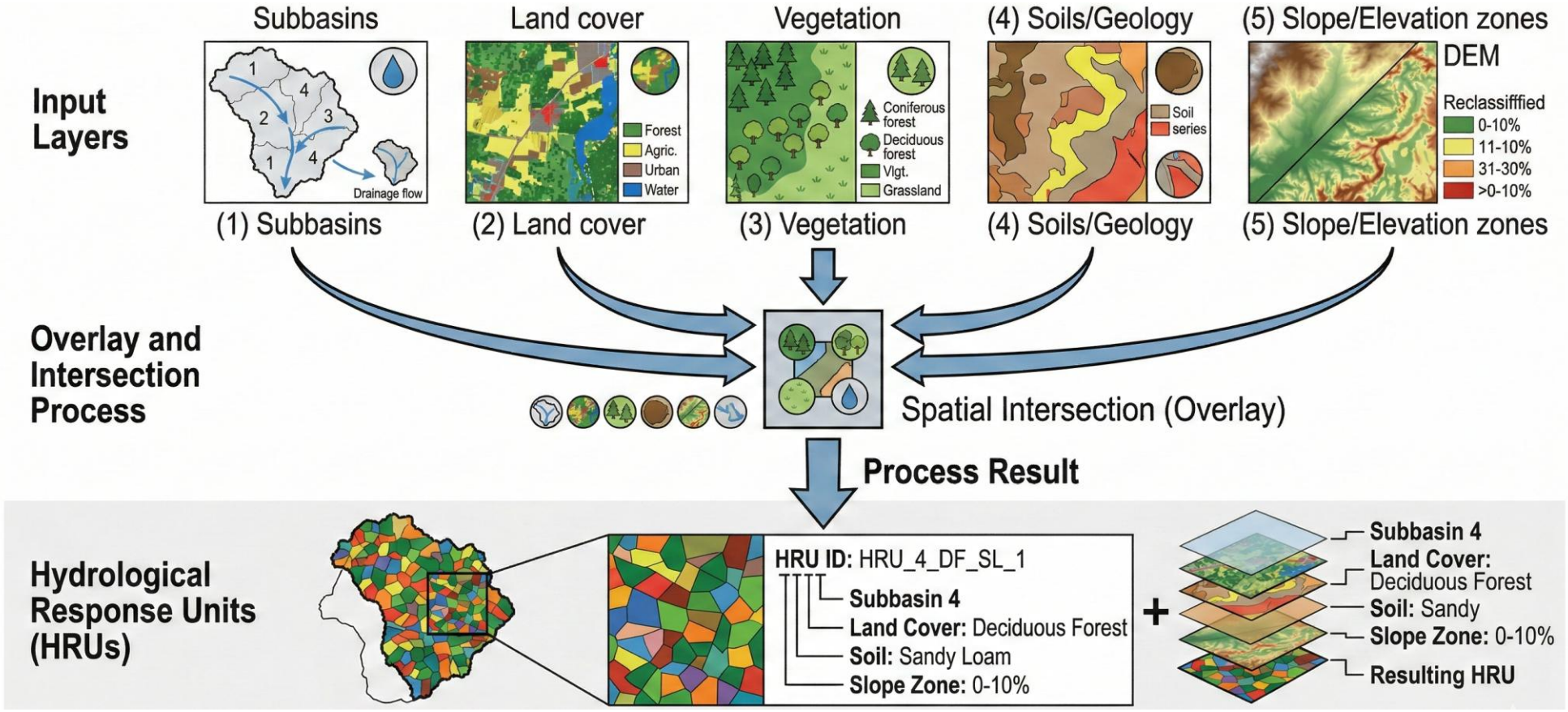
Raster overlay (grids)

- Tools: **Reclassify, Raster Calculator**
- Output: new categorical zones (HRU classes)

Overlay often increases feature count dramatically → plan for **QAQC and simplification**
Use a consistent **projected CRS** before overlay to avoid misalignment and incorrect areas

Core GIS Concepts

(6) Overlay concept



Core GIS Concepts

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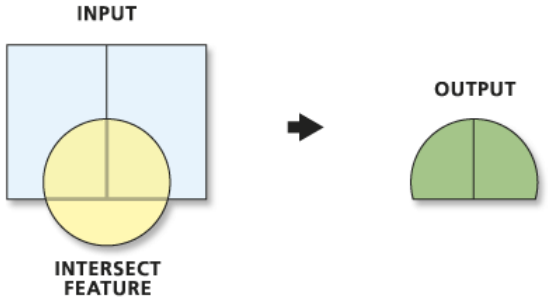
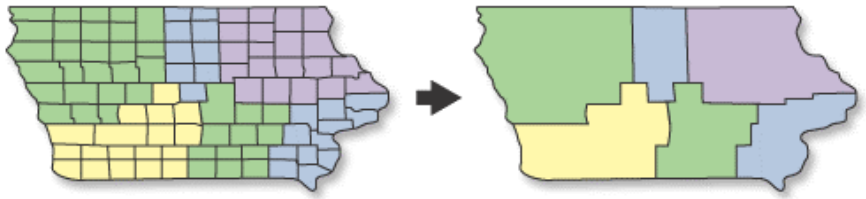
Key GIS Operations

General data processing operations

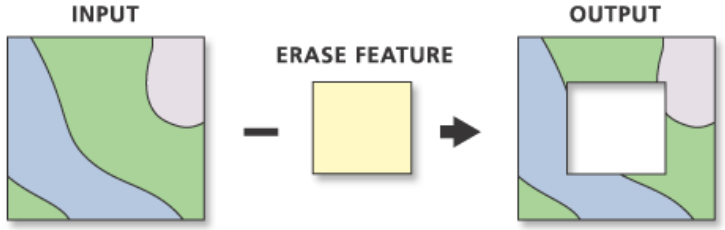
Intersect



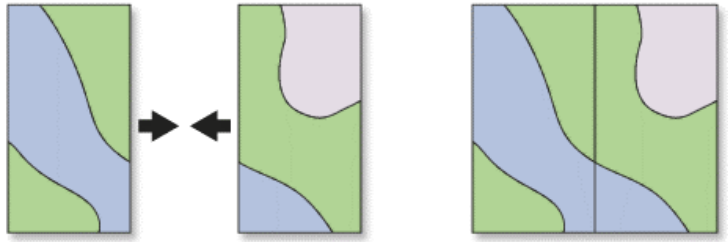
Dissolve



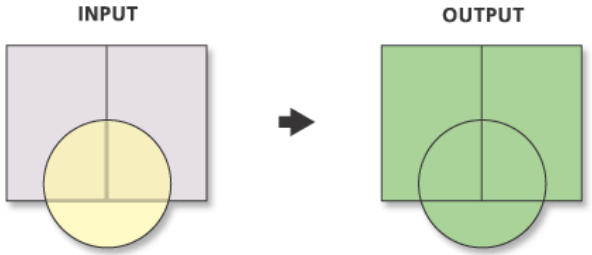
Erase



Merge



Union



Eliminate

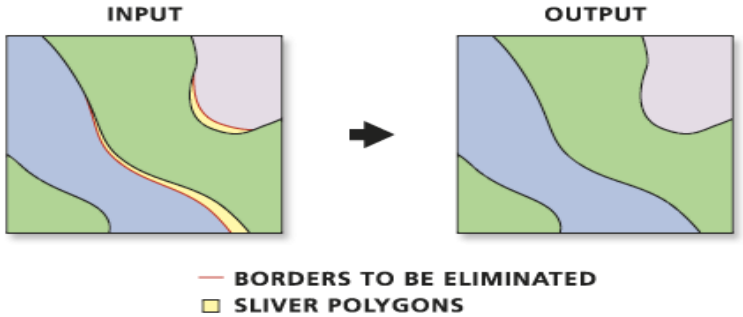


Table Join

NAME
San Francisco
Houston
Miami
Chicago
New York

NAME	NFL_Team
San Francisco	49ers
Houston	Texans
Miami	Dolphins
Chicago	Bears
New York	Jets

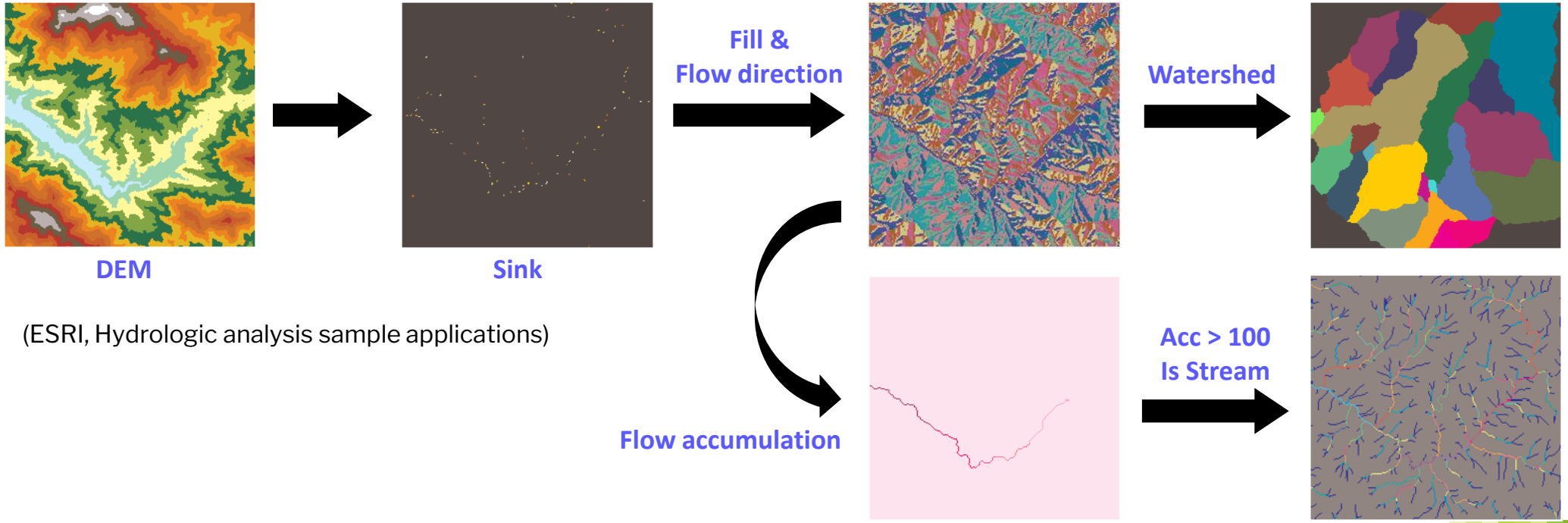
CITY	NFL_Team
San Francisco	49ers
Houston	Texans
Miami	Dolphins
Chicago	Bears
New York	Jets

Key GIS Operations for Hydrology

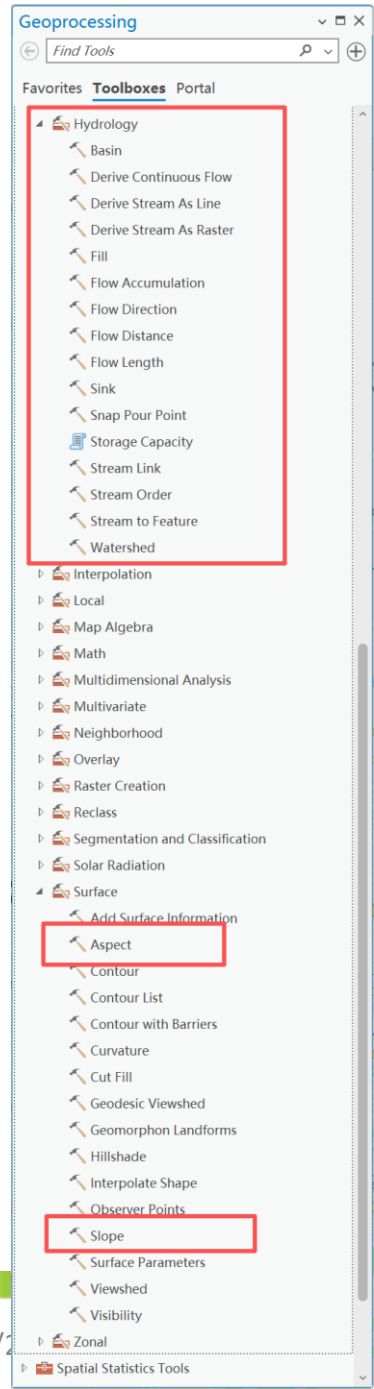
DEM-Based Hydrology Tools

Geoprocessing → Spatial Analyst Tools → **Hydrology**

Geoprocessing → Spatial Analyst Tools → **Surface**



(ESRI, Hydrologic analysis sample applications)



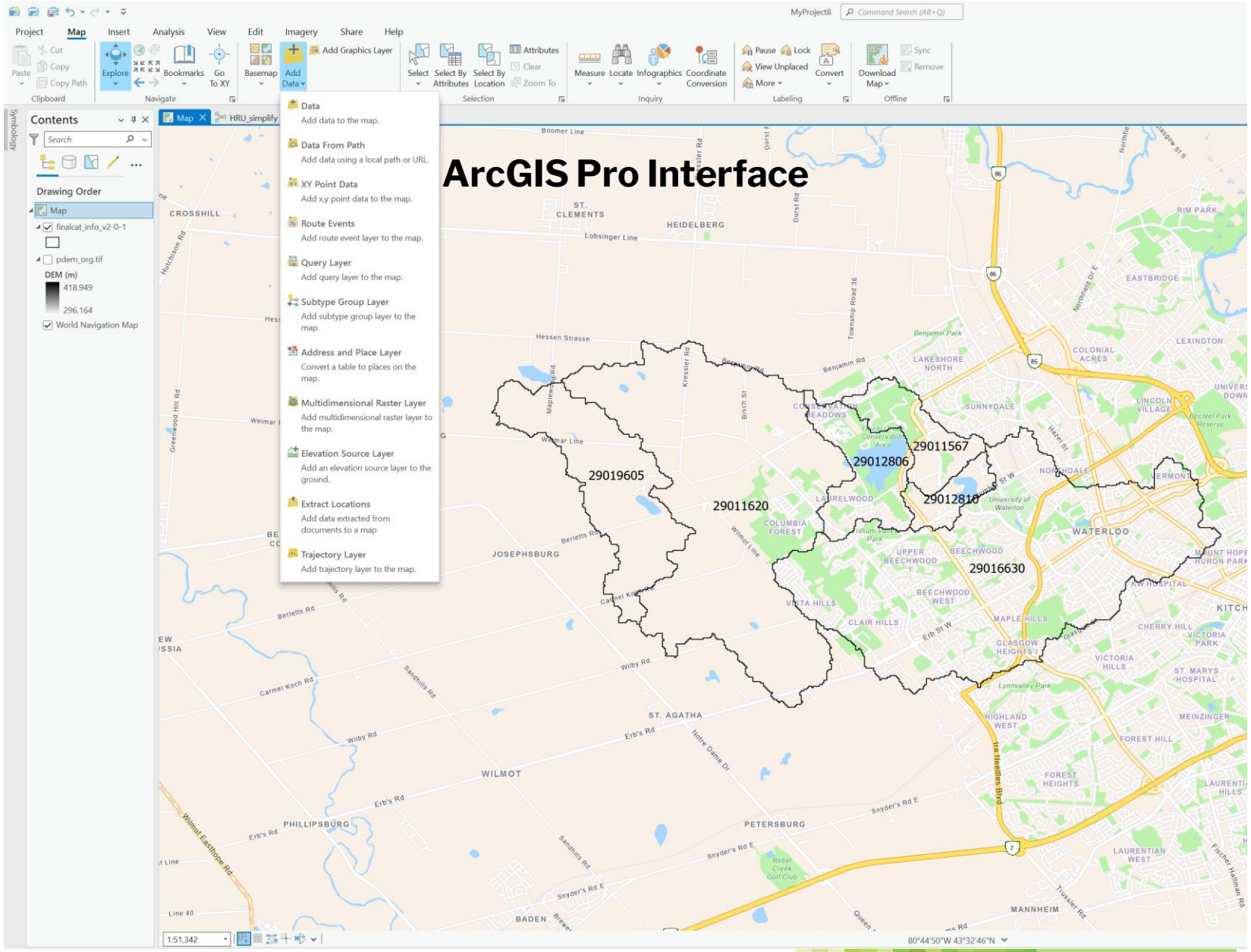
GIS Software

ArcGIS Pro

- **Desktop GIS application** developed by **Esri** (Environmental Systems Research Institute)
- Widely used in academia, government, and industry for mapping, spatial analysis, and geoprocessing
- Preferred for this course

QGIS

- Free, open-source desktop GIS application
- Alternative to ArcGIS Pro



ArcGIS Preview

- Manage and visualize vector + raster hydrology datasets (Rivers, lakes, subbasins, DEM, land cover, soils)
- Run geoprocessing workflows to build HRUs
- Extract tables and HRU attributes for modeling
- Ensure reproducibility and QAQC

Key panes we will use in the lab

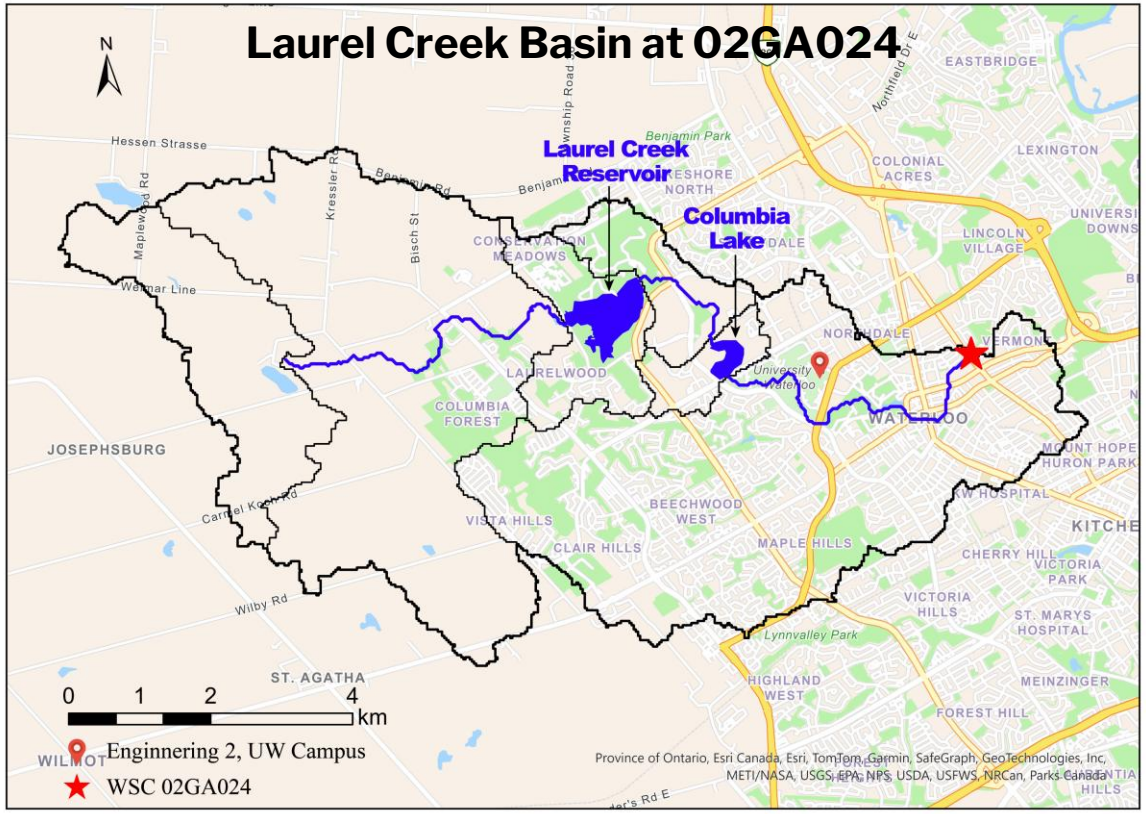
- **Catalog** pane: your data & geodatabases (recommended: File GDB)
- **Contents** pane: layer order, visibility, symbology
- **Map** view: explore and select features
- **Geoprocessing** pane: search tools, set parameters, run, view messages
- **Modelbuilder**: Use pre-built tool (HRU_simplify) to run a workflow in the lab
- **Attribute table**: inspect fields/IDs, select, summarize, export

Lab Preview: HRU Discretization Workflow

Refer details to the walkthrough document: **`/Docs/HRU_discretization_walkthrough.pdf`**

Our study area is the basin draining to the WSC streamflow gauging station 02GA024 (**LAUREL CREEK AT WATERLOO**).

- The total drainage area is 58 km^2. Only **6 subbasins**.
- **Two small lakes** are in the basin (Laurel Creek Reservoir and the Columbia Lake).
- Land cover is dominated by **urban and agriculture**.
- Hongren has prepared a soil profile map (polygons) for this basin, including **drained_agriculture_soil and default_soil**.
- You need to process land cover raster by **reclassifying** land cover classes, and then use it to **define a vegetation layer**.
- HRU = Subbasin × Lake ×Landcover × Veg × Soil
- We do not want sliver non-lake HRUs with area < 1 km^2) in modeling



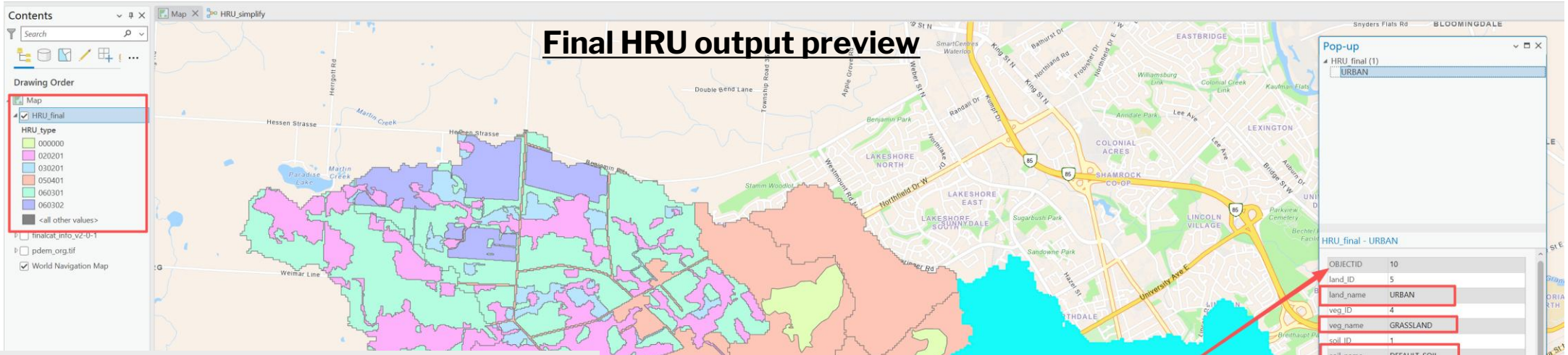
Assumption #1: Lake extent should stay as-is; not intersect with others.

Assumption #2: We need per-subbasin unique HRUs, i.e., if two same-type HRUs falling into subbasins A and B, they should be encoded differently (through HRU_ID).

Lab Preview: HRU Discretization Workflow

Refer details to the walkthrough document: **`/Docs/HRU_discretization_walkthrough.pdf`**

Assumption #1: Lake extent should stay as-is; not intersect with others. **Assumption #2:** We need per-subbasin unique HRUs (through HRU_ID).



HRU_final

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	OBJECTID *	Shape *	land_ID	land_name	veg_ID	veg_name	soil_ID	soil_name	HRU_type	SubId	HRU_area	HRU_ID
1	1	Polygon	5	URBAN	4	GRASSLAND	1	DEFAULT_SOIL	050401	29011567	2.5956	29011567050401
2	2	Polygon	2	WETLAND	2	FOREST	1	DEFAULT_SOIL	020201	29011620	3.190725	29011620020201
3	3	Polygon	3	FOREST	2	FOREST	1	DEFAULT_SOIL	030201	29011620	3.252825	29011620030201
4	4	Polygon	5	URBAN	4	GRASSLAND	1	DEFAULT_SOIL	050401	29011620	1.72935	29011620050401
5	5	Polygon	6	AGRICULTURE	3	AGRICULTURE	1	DEFAULT_SOIL	060301	29011620	10.414013	29011620060301
6	6	Polygon	6	AGRICULTURE	3	AGRICULTURE	2	DRAINED_AGRICULTU...	060302	29011620	1.984387	29011620060302
7	7	Polygon	5	URBAN	4	GRASSLAND	1	DEFAULT_SOIL	050401	29012806	2.895893	29012806050401
8	8	Polygon	5	URBAN	4	GRASSLAND	1	DEFAULT_SOIL	050401	29012810	1.15056	29012810050401
9	9	Polygon	3	FOREST	2	FOREST	1	DEFAULT_SOIL	030201	29016630	1.09395	29016630030201
10	10	Polygon	5	URBAN	4	GRASSLAND	1	DEFAULT_SOIL	050401	29016630	22.16205	29016630050401
11	11	Polygon	2	WETLAND	2	FOREST	1	DEFAULT_SOIL	020201	29019605	2.211975	29019605020201
12	12	Polygon	6	AGRICULTURE	3	AGRICULTURE	1	DEFAULT_SOIL	060301	29019605	4.711725	29019605060301
13	13	Polygon	0	LAKE	0	LAKE	0	LAKE	000000	29012806	0.627607	29012806000000
14	14	Polygon	0	LAKE	0	LAKE	0	LAKE	000000	29012810	0.16974	29012810000000

Lab Preview: HRU Discretization Workflow

Refer details to the walkthrough document: **`/Docs/HRU_discretization_walkthrough.pdf`**

Assumption #1: Lake extent should stay as-is; not intersect with others. **Assumption #2:** We need per-subbasin unique HRUs (through HRU_ID).

Final HRU output preview

After simplification, there are 6 HRU types in `HRU_final` layer.

land_name	veg_name	soil_name
AGRICULTURE	AGRICULTURE	DEFAULT_SOIL
AGRICULTURE	AGRICULTURE	DRAINED_AGRICULTURE_SOIL
FOREST	FOREST	DEFAULT_SOIL
URBAN	GRASSLAND	DEFAULT_SOIL
WETLAND	FOREST	DEFAULT_SOIL
LAKE	LAKE	LAKE

Finally, we need to export HRU attributes to **02GA024.rvh** file.

02GA024.rvh

C: > Users > H53SHEN > Desktop > GIS_lecture_2026 > ENVE 383_GIS_materials > data_benchmark > v2-0-1a > 02GA024.rvh

```
:SubBasins
  29019605        ODI_804             29011620          Chn_ZERO_LENGTH     ZERO-           0
:EndSubBasins

:SubBasinProperties
  :Parameters     Q_REFERENCE     TIME_CONC     TIME_TO_PEAK
  :Units          none            none          none
  29011567        5.325830        0.045500      0.013650
  29011620        4.356320        0.099918      0.029975
  29012806        4.914580        0.051104      0.015331
  29012810        5.535020        0.035193      0.010558
  29016630        9.219710        0.104685      0.031406
  29019605        1.096990        0.066059      0.019818
:EndSubBasinProperties

:HRUs
  :Attributes       AREA              ELEVATION     LATITUDE     LONGITUDE     BASIN_ID     LAND_USE_CLASS     VEG_CLASS     SOIL_PROFILE                AQUIFER_PROFILE     TERRAIN_CLASS     SLOPE       ASPECT
  :Units            km2               m             deg          deg           none         none               none          none                        none                none              deg         deg
  29011567050401    2.5956000000      345.93        43.483932    -80.567688    29011567     URBAN              GRASSLAND     DEFAULT_SOIL                [NONE]              [NONE]            2.068323    153.404199
  29011620020201    3.1907250000      369.81        43.479565    -80.613211    29011620     WETLAND            FOREST        DEFAULT_SOIL                [NONE]              [NONE]            3.262719    166.917507
  29011620030201    3.2528250000      369.81        43.470164    -80.617634    29011620     FOREST             FOREST        DEFAULT_SOIL                [NONE]              [NONE]            3.262719    166.917507
  29011620050401    1.7293500000      369.81        43.473755    -80.603662    29011620     URBAN              GRASSLAND     DEFAULT_SOIL                [NONE]              [NONE]            3.262719    166.917507
  29011620060301    10.4140125000     369.81        43.468539    -80.613626    29011620     AGRICULTURE        AGRICULTURE   DEFAULT_SOIL                [NONE]              [NONE]            3.262719    166.917507
  29011620060302    1.9843875000      369.81        43.493647    -80.628010    29011620     AGRICULTURE        AGRICULTURE   DRAINED_AGRICULTURE_SOIL    [NONE]              [NONE]            3.262719    166.917507
  29012806050401    2.8958933884      349.24        43.477448    -80.582251    29012806     URBAN              GRASSLAND     DEFAULT_SOIL                [NONE]              [NONE]            1.757881    160.431606
  29012806000000    0.6276066163      349.24        43.479611    -80.576567    29012806     LAKE               LAKE          LAKE                        [NONE]              [NONE]            1.757881    160.431606
  29012810050401    1.1505603664      341.40        43.472323    -80.561044    29012810     URBAN              GRASSLAND     DEFAULT_SOIL                [NONE]              [NONE]            1.747721    161.970201
  29012810000000    0.1697396368      341.40        43.473466    -80.556412    29012810     LAKE               LAKE          LAKE                        [NONE]              [NONE]            1.747721    161.970201
  29016630030201    1.0939500000      352.34        43.458178    -80.570019    29016630     FOREST             FOREST        DEFAULT_SOIL                [NONE]              [NONE]            2.289449    162.468970
  29016630050401    22.1620500000     352.34        43.459138    -80.548814    29016630     URBAN              GRASSLAND     DEFAULT_SOIL                [NONE]              [NONE]            2.289449    162.468970
  29019605020201    2.2119750000      376.80        43.482090    -80.645742    29019605     WETLAND            FOREST        DEFAULT_SOIL                [NONE]              [NONE]            3.490921    167.239563
  29019605060301    4.7117250000      376.80        43.476088    -80.643971    29019605     AGRICULTURE        AGRICULTURE   DEFAULT_SOIL                [NONE]              [NONE]            3.490921    167.239563
:EndHRUs

:RedirectToFile  Lakes.rvh

# ------------------------------------------------------------
# Subbasin Group Definitions
# Allsubbasins: Subbasins utilized in calibration. Valid for TIME_CONC and TIME_TO_PEAK parameters.
# RiverChannelSubbasins: Subbasins with non-zero river length.
# LakeSubbasin_at_<Gauge> and LakeSubbasin_at_<SubId> are exclusive with LakeSubbasin_Ungauged.
# ------------------------------------------------------------

:SubBasinGroup Allsubbasins
  29011567  29011620  29012806  29012810  29016630  29019605
:EndSubBasinGroup

:SubBasinGroup RiverChannelSubbasins
  29011567  29011620  29016630
:EndSubBasinGroup
```

Lab Preview: HRU Discretization Workflow

If you store your ArcGIS working folder in OneDrive, make sure OneDrive syncing is turned OFF when running this lab!

Before You Start (Checklist)

(1) Download the lab package from LEARN

- Download and unzip: `ENVE 383_GIS_materials.zip`

(2) Read the walkthrough (Sections 1–2)

- Document: `/Docs/HRU_discretization_walkthrough.pdf`
- Make sure required software is installed (or plan to use the lab desktop computers).

(3) Check the provided datasets in ArcGIS Pro

- Load each dataset into an ArcGIS Pro map
- Check CRS, resolution (rasters), extent, and attribute tables (vectors)

(4) Create and organize your ArcGIS Pro project

- Create a new ArcGIS Pro project (you'll be prompted the first time you open ArcGIS Pro)
- Save the project in: `/working_folder/` (recommended)
- Keep all lab outputs in this folder (ArcGIS Pro will create a project geodatabase here as well)

Notes for This Lab

- You may **skip Section 4.4** in the walkthrough. After completing **Section 4.3**, you can jump directly to **Section 4.5**.
- **HRU simplification (Section 4.4)** can take up to ~1 hour, so it is an **optional** exercise.
- If you skip simplification, use the benchmark HRU layer provided here: **`/data_benchmark/v2-0-1a/HRU_final.shp` (for preparing the Raven HRU file)**

Lab Preview: HRU Discretization Workflow

After You Created HRU layer (`HRU_raw_join`)

(1) Populate attributes to the HRU layer

- Beyond the HRU type-related attributes, we need more terrain attributes for Raven
- Follow **Section 4.5** to calculate those attributes. Note that we use attributes from subbasin layer as an approximation. Thus, you do not need to calculate them from a DEM.

(2) Manual QAQC is required

- Section 4.6 listed QAQC approaches
- Follow those QAQC guidelines, check your data. If you are unsure about any results, ask for help!

(3) Export HRU feature class to a shapefile and a GeoJSON

- Section 4.7 introduces these two types of files.

(4) Translate HRU attributes into Raven RVH file

- If you skipped Section 4.4, make sure you load **`/data_benchmark/v2-0-1a/HRU_final.shp`** for this step
- Export attribute table to a CSV file, and then edit columns in Excel.
- You are encouraged to read the **Raven Manual Page 206-207** to learn about the HRU block in RVH file.

Lab Preview: HRU Discretization Workflow

Hongren will demo **ArcGIS Pro basics** & **key operations** during this lab…

We will record Hongren's screen during this demo and share the video with you.

Thank you!

Questions are welcome!

Email: hongren.shen@uwaterloo.ca